

Metrological Domain Profiling III: Reconstructing the 3D Tactile Grammar of the Inca Khipu

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Abstract

For over a century, computational analyses of the Inca khipu have been constrained by what we term the "Spreadsheet Fallacy" -- the assumption that the khipu operates primarily as a flat, two-dimensional, base-10 arithmetic ledger. This model fails to account for the fact that only 4.6% of known cord clusters demonstrate valid summation. In this paper, we extend Metrological Domain Profiling (MDP) to analyse 54,403 cords across 619 khipus from the Open Khipu Repository, moving beyond one-dimensional colour profiling to reconstruct the full three-dimensional, tactile, and hierarchical ontology of the system. We demonstrate that the khipu possesses strict spatial and material structure operating across four distinct layers: (1) **Material Metrology**, where fiber type (cotton vs camelid) redefines numerical scale by up to 67x; (2) **Topological Syntax**, where administrative granularity is encoded in subsidiary cord depth and colour palette shifts systematically with hierarchical level; (3) **Categorical Syntax**, featuring statistically constrained colour sequences ($p < 0.001$) that demonstrate strict institutional sorting rules rather than random clustering, with same-colour run lengths spiking at decimal administrative units; and (4) **Hardware Metadata**, where physical features including canutito thread-wrappings (98.6% colour-independent from parent cords), primary cord construction, and cord termination types encode document-level metadata and institutional information. These findings suggest the khipu is not merely a mathematical ledger, but a multi-layered, tactile administrative system whose information is distributed across the material, spatial, and structural dimensions of the textile.

Data: Open Khipu Repository, DOI: 10.5281/zenodo.18025748

Prior work: MDP Paper I (DOI: 10.17613/mmjdt-ba806), Paper II (DOI: 10.17613/q536f-0wa13)

Code: github.com/souldriver007/mdp-ancient-scripts

1. Introduction

1.1 The Spreadsheet Fallacy

The Inca khipu is a knotted-cord recording device used throughout the Tawantinsuyu (Inca Empire, c. 1400-1532 CE) for administrative record-keeping. While leading theorists -- notably Ascher & Ascher (1981) and Urton (2003) -- have long argued for the three-dimensional, non-numerical complexity of the khipu, including Urton's proposed 7-bit binary code based on spin, ply, and knot direction, computational and statistical models have continually defaulted to what we term the "Spreadsheet Fallacy" -- the attempt to

computationally validate khipus primarily as flat, base-10 arithmetic ledgers. Under this dominant computational paradigm, pendant cords carry numerical values, cord colours distinguish commodity categories, and clusters of cords should sum to produce totals.

This model is not wrong -- it is incomplete. Our analysis of 619 khipus confirms that only 4.6% of cord clusters pass a flat summation test. When we test summation hierarchically (parent cord = sum of subsidiary cords), the rate improves to 17.4% -- still leaving more than 80% of the corpus unexplained by arithmetic alone. The spreadsheet model captures at most one layer of what is clearly a multi-layered system.

1.2 From 1D to 3D

In MDP Papers I and II (Sharman 2026a, 2026b), we demonstrated that khipu cord colours cluster into distinct numerical domains -- the same Administrative Segregation Principle (ASP) found in Proto-Cuneiform, Proto-Elamite, Indus Valley, Linear A, Linear B, and Cretan Hieroglyphic scripts across four continents and 4,600 years. However, those papers treated the khipu primarily as a one-dimensional sequence of coloured cords with knots.

The khipu is not a 1D sequence. It is a 3D tactile object with subsidiary tree structures (cords hanging from other cords), physical properties detectable by touch in darkness (fiber material, twist direction), spatial positioning (attachment points along the primary cord), and secondary metadata layers (canutito thread wrappings, termination types, primary cord construction). This paper systematically profiles these variables across the complete Open Khipu Repository. While archaeologists have observed features like colour banding, material differences, and subsidiary structures on individual khipus, MDP provides the first computational framework to statistically validate their administrative functions across a 54,403-cord corpus, quantify their interactions, and test specific structural hypotheses against randomised controls.

1.3 Methodology

All analysis was performed on the Open Khipu Repository SQLite database (619 khipus, 54,403 cords, 110,677 knots; DOI: 10.5281/zenodo.18025748). Cord values were computed from knot records following the standard Ascher decoding. Statistical significance was assessed via chi-squared tests for categorical distributions and permutation tests with 10,000 iterations for continuous value comparisons. Key findings were verified via independent raw SQL queries against the database.

2. Material Metrology: The Medium Is the Math

2.1 Fiber Type Redefines Colour Domains

The same colour on different fiber materials carries dramatically different numerical values. Cotton, produced on the Pacific coast, and camelid fiber (alpaca/llama), produced in the highland Andes, would have been immediately distinguishable by touch. The biology of the string is part of the metrology.

Colour	Cotton Mean	Camelid Mean	Ratio
AB (light brown)	294	7	41x
W (white)	409	25	17x
MB (moderate brown)	153	10	15x
B (mod. yellowish brown)	93	5	19x
FB (brownish black)	718	11	67x

This finding was verified via independent raw SQL computation. Cotton AB (n=5,265) has mean=289, Camelid AB (n=90) has mean=7.

2.2 The Black Cord (LK) Anomaly

Black cords use a fundamentally inverted knot syntax. Where all other colours show approximately 68.5% S-type (single) knots and 23.0% L-type (long) knots, black cords show 79.5% L-type and 16.3% S-type -- a near-perfect inversion confirmed at chi-squared = 2797.7, df=2, $p < 0.001$. Additional convergent properties:

- 91.3% appear in MIDDLE position within clusters; only 5.4% FIRST, 3.2% LAST
- 0.31x depleted at subsidiary depths -- almost exclusively top-level entries
- 0.18x among knotless cords -- black cords almost never appear without knots
- 62% of all canutito-bearing cords are black (see Section 5.1)

Taken together, these properties define black cords as a structurally segregated mathematical domain. They are not "another commodity colour" -- they are a functionally distinct accounting instrument. Their consistent avoidance of first and last positions suggests LK cords function as deeply nested modifiers or values bound to adjacent structural cords, rather than independent entries.

2.3 Document Rank via Primary Cord Construction

The physical beginning of the primary cord correlates with administrative scale:

Beginning Type	Khipus	Median Total Value
Needlework bundle	8	26,247
Wooden bar	2	6,666
Tassel/borla	9	3,501
Knotted	72	1,531
Doubled	72	1,344
Twisted	284	561

Khipus categorised by maximum cord value show clean monotonic physical scaling: Chunka (≤ 10) khipus average 35 cords and 43.6cm primary cords; Waranqa (101-1,000) khipus average 108 cords and 67.2cm. The physical size of the document directly encodes its bureaucratic tier.

3. Topological Syntax: The 3D Tree

3.1 Depth Encodes Administrative Granularity

Depth	Cords	Median Value	Mean Value
0 (pendants)	24,670	15	326
1 (subsidiary L1)	8,970	4	69
2 (subsidiary L2)	1,153	6	29
3 (subsidiary L3)	172	10	38

Values decrease sharply with each level of depth. KB (dark brown) is 2.19x enriched at subsidiary depths. BG (grayish blue) is 2.23x enriched. Meanwhile, W (white) drops from 28.6% to 18.9%, and LK (black) is 0.31x depleted. The colour palette shifts with depth, suggesting different administrative functions at each hierarchical level.

3.2 Top Cords: A Third Administrative Domain

232 cords hang upward from the primary cord (CORD_LEVEL < 0). These "top cords" form a distinct domain: median value = 48, mean = 122 (between pendants at mean=326 and subsidiaries at mean=69), with an MB-dominated colour palette and 85% S-type knots. They represent a third administrative domain, possibly institutional headers or routing labels.

3.3 Subsidiary Summation

Testing parent-child summation (does a pendant's value equal the sum of its subsidiaries' values?) yields a 17.4% combined match rate -- 3.8x better than the traditional flat cluster summation test at 4.6%. The median parent/children ratio is 1.57, meaning parents are typically 57% larger than the sum of their children -- suggesting an additional component such as tax, overhead, or reserve allocation.

3.4 First Pendant as Document Header

The first pendant cord on each khipu carries 5.7x the mean value of body pendants (mean=1,716 vs mean=303). Additionally, 34% of first pendants have no knots at all. This parallels heading signs found in Proto-Elamite (M157) and Linear A (A-KA-RU), suggesting a header function -- either a summary total or an institutional identifier.

4. Categorical Syntax: Administrative Sorting in String

4.1 Constrained Colour Transitions

We constructed a Markov transition matrix for cord colour sequences along the primary cord. Self-transitions dominate massively: DB to DB at 34.0x expected, NB to NB at 23.7x. Certain cross-colour transitions are completely absent: B to NB has 0 occurrences against 38 expected. NB to B has 0 against 38. LB to AB has 1 against 193.

Transition	Observed	Expected	Ratio
DB to DB	466	14	34.0x
NB to NB	317	13	23.7x
LB to LB	675	32	21.1x
B to NB	0	38	0.00x (forbidden)
NB to B	0	38	0.00x (forbidden)
LB to AB	1	193	0.01x (avoided)

While this statistical rigidity superficially mimics linguistic grammar, in an administrative context it more likely represents institutional sorting protocols -- for example, a rule that commodity X must always be recorded before commodity Y. Whether these constraints reflect administrative convention, institutional regulation, or deeper structural properties remains an open question.

4.2 Run-Length and Decimal Administrative Units

Same-colour runs spike at specific lengths: runs of exactly 10 are 3.55x over-represented relative to geometric baseline, and runs of 20 are 160x over-represented. Runs of 5 are under-represented at 0.44x. The over-representation at 10 and 20 aligns with the Inca decimal administrative hierarchy (Chunka = 10 households).

4.3 Functional Taxonomy: Structural Markers vs Data Carriers

By profiling the 11,801 pendant cords that carry no knots, we discovered a functional taxonomy. KB (dark brown) is 2.14x enriched among knotless cords -- a structural marker. LK (black) is 0.18x -- almost never appears without knots, a pure data carrier. This creates a binary classification of cord colours by administrative function.

4.4 Barberpole Encoding as Morphological Modification

Compound-colour cords (barberpole pattern) carry systematically lower values than solid equivalents: solid W mean=403, barberpole W mean=148 (63% reduction). Solid AB mean=315, barberpole AB mean=127 (60% reduction). This parallels compound determinatives in Proto-Elamite, suggesting barberpole encodes sub-categories.

5. Hardware Metadata: Physical Encoding

5.1 Canutito: An Independent Metadata Layer

461 cords have canutito data -- thread wrappings with independently recorded colours. 98.6% of canutito colours differ from their parent cord colour (only 6 of 419 matched). 62% of canutito parent cords are black (LK). The canutito palette includes vivid reds, blues, and deep browns rare in the main cord palette. This near-total independence proves the canutito is a separate information channel -- the cord colour encodes the category, the knots encode the value, and the canutito wrapping encodes a third dimension of information whose specific meaning remains to be determined.

5.2 Axis Orientation

24% of knots have recorded axis orientation: AXD (axis down, 10,888) and AXU (axis up, 4,181). AXD-dominant cords carry higher mean values (371) than AXU-dominant (154). This variable warrants further investigation but the 76% missing data rate limits current conclusions.

6. Negative Findings

Three hypotheses were tested and falsified. We report these explicitly to constrain the space of viable ontological models.

- **Cluster spacing as punctuation:** Physical gaps between cord clusters show no correlation with domain shifts. Colour shift rates after gaps exceeding 5cm (46.1%) are indistinguishable from rates after smaller gaps (45.1%).
- **Hanan/Hurin midpoint split:** Splitting khipus at the physical midpoint reveals no structural asymmetry in colour, twist, or value distributions between left and right halves.
- **Cord thickness as domain marker:** No monotonic relationship between cord thickness and value magnitude was found.

7. Discussion

7.1 The Multi-Channel Ontology

Our findings suggest the khipu encodes information across at least six independent channels: (1) cord colour for commodity identification; (2) knot type and position for numerical value; (3) fiber material for administrative domain and scale; (4) tree depth for hierarchical granularity; (5) physical construction for document-level metadata; and (6) canutito wrapping as an independent metadata layer. Each channel operates independently -- canutito colours bear no relationship to cord colours, fiber modifies colour domains by up to

67x, and tree depth reshuffles the colour palette.

7.2 Beyond the Spreadsheet

The Spreadsheet Fallacy is not that arithmetic is absent from khipus -- our subsidiary summation test confirms it operates in at least 17.4% of parent-child relationships. The fallacy is the computational assumption that arithmetic is the primary or sole organisational principle. The colour transition constraints, run-length distributions, and functional taxonomy collectively demonstrate that the khipu follows strict categorical sorting rules beyond simple arithmetic.

The Inca Empire managed the largest contiguous landmass in the pre-Columbian Americas without a written alphabet -- not because they lacked the capacity for complex information encoding, but because they distributed their administrative information across the topological, material, and tactile dimensions of textiles.

7.3 Site-Level Variation

Preliminary comparison between Pachacamac (90 khipus, 5,916 cords) and Cieneguilla, Valle de Lurin (2 khipus, 21 cords) reveals divergent MDP profiles. Cieneguilla shows 75% L-type knots versus Pachacamac's 17%. With only 21 cords, this is suggestive rather than conclusive. Validation requires application to larger corpora from these sites, including collections not yet represented in the OKR.

7.4 Limitations

Fiber type is recorded for only 58% of cords. Axis orientation is recorded for 24% of knots. The OKR's provenance data is unevenly distributed. The canutito dataset (461 cords) is small relative to the full corpus. All findings should be understood as corpus-scale statistical tendencies rather than universal rules.

8. Conclusion

This paper extends Metrological Domain Profiling from one-dimensional sign-numeral analysis to three-dimensional tactile-spatial analysis, demonstrating that the Inca khipu encodes administrative information across at least six independent channels: colour, knots, fiber material, tree depth, physical construction, and canutito wrapping. The discovery of constrained colour transitions, decimal run-length groupings, a functional taxonomy of structural versus data-carrying cord colours, and an independent canutito metadata layer collectively argue against the Spreadsheet Fallacy and in favour of understanding the khipu as a multi-layered administrative system whose information is distributed across the physical dimensions of the textile.

Methods Note

All analysis scripts (`khipu_mdp_v3_5_final.py`, `khipu_verify.py`, `khipu_audit.py`) are available in the project repository alongside the earlier MDP scripts for Proto-Cuneiform, Proto-Elamite, Indus Valley, Linear A, Linear B, and Cretan Hieroglyphic analysis.

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